

Intro to ML Engineering and Linear Regression:

Pre-reads

1. Introduction

Description:

Machine Learning (ML) is a branch of Artificial Intelligence (AI) that enables systems to learn patterns from data and make decisions or predictions without being explicitly programmed.

Importance:

- Automation of decision-making.
 - Unlocks insights from complex data.
 - Powers modern applications like recommendation systems, NLP, and computer vision.
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2. ML vs Classical Programming

Aspect	Classical Programming	Machine Learning
Input	Program + Data	Data + Expected Outcome (labels)
Process	Explicit rules	Learn patterns from data
Output	Answers	Predictions / decisions

Importance: ML is key when rules are unknown or too complex to code manually.

3. SDE Pipeline

Description: Software Development Engineer (SDE) pipeline refers to the lifecycle of developing and deploying software.

Stages: Requirement → Design → Implementation → Testing → Deployment → Maintenance

Importance: Understanding the SDE pipeline helps in appreciating the differences when working with ML pipelines, which are data-driven rather than purely code-driven.

4. ML Pipeline

Description: ML pipeline is the end-to-end process of building, deploying, and monitoring ML models.

Typical Stages:

1. Problem definition
2. Data collection
3. Data cleaning & preprocessing
4. Feature engineering
5. Model selection & training
6. Evaluation & validation
7. Deployment & monitoring

Importance: Ensures reproducibility, scalability, and maintainability in ML projects.

5. Phases of Machine Learning

1. **Data Collection:** Gather relevant data.
2. **Data Preparation:** Clean and preprocess data.
3. **Modeling:** Choose algorithms and train models.
4. **Evaluation:** Measure model performance.
5. **Deployment:** Integrate model into applications.
6. **Monitoring & Maintenance:** Track performance over time.

Importance: Structured approach reduces errors and improves model reliability.

6. ML Categorization

Machine Learning can be categorized based on the type of supervision, data, and tasks.

7. Types of Learning

1. **Supervised Learning:** Learns from labeled data (input → output mapping).
2. **Unsupervised Learning:** Learns patterns from unlabeled data (clustering, dimensionality reduction).
3. **Reinforcement Learning:** Learns via rewards and penalties from interaction with the environment.

Importance: Helps in choosing the right approach for a given problem.

8. Assessments in ML

Types of Tasks:

- **Regression:** Predict continuous values (e.g., house price).
- **Classification:** Predict discrete labels (e.g., spam vs ham).
- **Clustering:** Group similar data points (e.g., customer segmentation).
- **Recommendation:** Suggest items based on user behavior (e.g., Netflix recommendations).
- **Time Series Forecasting:** Predict future values based on past trends (e.g., stock prices).
- **Reinforcement Learning Tasks:** Optimize decisions over time (e.g., game AI).

Importance: Knowing task types ensures the correct choice of algorithm and evaluation metric.

9. Linear Regression

Description: Predicts a continuous target using a linear relationship between features and target.

Formula: $y = w_1x_1 + w_2x_2 + \dots + b$ or $y = w_1x_1 + w_2x_2 + \dots + by = w_1x_1 + w_2x_2 + \dots + b$

Importance:

- Simple and interpretable.
 - Forms the foundation for understanding more complex models.
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10. Problem Statement Introduction

Description: A problem statement clearly defines the objective, data, and expected output.

Importance: Guides the choice of algorithms and evaluation metrics.

11. Data Notation

- X : Features / input variables
- y : Target variable / label
- m : Number of samples
- n : Number of features

Importance: Standard notation aids in understanding equations and algorithms.

12. Geometric Intuition

Description: ML concepts like linear regression or classification can be visualized geometrically (lines, planes, distances).

Importance: Helps in intuitive understanding of how models separate, predict, or cluster data.